

DATA 607 Assignment 05A CodeBase

Marcin K

2026-10-01

—- Part 1

- (1) Create a .CSV file (or optionally, a MySQL database!) that includes all of the information above. You're encouraged to use a “wide” structure similar to how the information appears above, so that you can practice tidying and transformations as described below

Transformed data from wide to long format. (+10)

-
1. Created a .csv file in the exact same format as the assignment.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2     4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr       1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)
```

```
## Warning: package 'scales' was built under R version 4.6.1
```

```
##
## Attaching package: 'scales'
##
```

```
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(knitr)

data_path <- file.choose()

data <- read_csv(data_path)
```

```
## New names:
## Rows: 5 Columns: 7
## -- Column specification
## ----- Delimiter: "," chr
## (2): ...1, ...2 dbl (5): Los Angeles, Phoenix, San Diego, San Francisco,
## Seattle
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'
## * ' -> '...2'
```

```
data2 <- data %>% pivot_longer(
  cols = !c(1, 2),
  names_to = "Cities",
  values_to = "Arrivals"
)

data2[6:10,1] <- "ALASKA"
data2[21:25,1] <- "AM WEST"
colnames(data2)[1:2] <- c("Airline", "Status")

data3 <- filter(data2, !is.na(Airline))

data3 %>% print(n=20)
```

```
## # A tibble: 20 x 4
##   Airline Status Cities      Arrivals
##   <chr>    <chr>  <chr>      <dbl>
## 1 ALASKA  on time  Los Angeles    497
## 2 ALASKA  on time  Phoenix        221
## 3 ALASKA  on time  San Diego      212
## 4 ALASKA  on time  San Francisco  503
## 5 ALASKA  on time  Seattle      1841
## 6 ALASKA  delayed  Los Angeles     62
## 7 ALASKA  delayed  Phoenix         12
## 8 ALASKA  delayed  San Diego       20
## 9 ALASKA  delayed  San Francisco  102
## 10 ALASKA delayed  Seattle       305
## 11 AM WEST on time  Los Angeles    694
```

## 12 AM WEST on time Phoenix	4840
## 13 AM WEST on time San Diego	383
## 14 AM WEST on time San Francisco	320
## 15 AM WEST on time Seattle	201
## 16 AM WEST delayed Los Angeles	117
## 17 AM WEST delayed Phoenix	415
## 18 AM WEST delayed San Diego	65
## 19 AM WEST delayed San Francisco	129
## 20 AM WEST delayed Seattle	61

—- Part 2

- (2) Read the information from your .CSV file into R, and use tidyr and dplyr as needed to tidy and transform your data

Submitted data file with at least some count analysis (+45)

Recreated File in Same Format as given, including missing data where there were empty cells in source data. (+5)

Provided code to populate missing data. (+5)

The section shows what I assume are either on-time or delayed arrivals/departures at five U.S. airports from carriers Alaska and AM West (now part of American Airlines). For the sake of this assignment, we will assume they are arrivals.

```
# Chunk 3 satisfies the count portion of the assignment.

AL_ontime <- data3 %>% filter(Status == "on time" & Airline == "ALASKA") %>% pull(Arrivals) %>% sum()
AMW_ontime <- data3 %>% filter(Status == "on time" & Airline == "AM WEST") %>% pull(Arrivals) %>% sum()
AL_delay <- data3 %>% filter(Status == "delayed" & Airline == "ALASKA") %>% pull(Arrivals) %>% sum()
AMW_delay <- data3 %>% filter(Status == "delayed" & Airline == "AM WEST") %>% pull(Arrivals) %>% sum()
AL_total <- data3 %>% filter(Airline == "ALASKA") %>% pull(Arrivals) %>% sum()
AMW_total <- data3 %>% filter(Airline == "AM WEST") %>% pull(Arrivals) %>% sum()

print(paste0(
  "Alaska Airlines saw an average delay percentage of ",
  percent(AL_delay / AL_total),
  " in flight arrivals whereas AM WEST saw a delay of ",
```

```
percent(AMW_delay / AMW_total),
" across all flights on the same five routes."
))
```

```
## [1] "Alaska Airlines saw an average delay percentage of 13% in flight arrivals whereas AM WEST saw a
```

```
print(paste0(
  "Alaska Airlines saw ",
  AL_delay,
  " delayed flights and ",
  AL_ontime,
  " on-time flights for a grand total of ",
  AL_total,
  " flights.",
  "AM West saw ",
  AMW_delay,
  " delayed flights and ",
  AMW_ontime,
  " on-time flights for a grand total of ",
  AMW_total,
  " flights."
))
```

```
## [1] "Alaska Airlines saw 501 delayed flights and 3274 on-time flights for a grand total of 3775 flight
```

```
# Chunk 4 transforms the long dataframe back to a wide dataframe.
```

```
data4 <- data3 %>%
  pivot_wider(
    names_from = Cities,
    values_from = Arrivals
  )

data4[2,1] <- ""
data4[4,1] <- ""
colnames(data4)[1:2] <- c("", "")

data4 <- data4 %>% add_row(.before = 3)
```

```
## New names:
## * ' ' -> '...1'
## * ' ' -> '...2'
```

```
data4
```

```
## # A tibble: 5 x 7
##   ...1      ...2   'Los Angeles' Phoenix 'San Diego' 'San Francisco' Seattle
##   <chr>    <chr>         <dbl>    <dbl>      <dbl>      <dbl>    <dbl>
## 1 "ALASKA" on time          497     221        212         503     1841
## 2 ""      delayed           62      12         20         102     305
## 3 <NA>    <NA>              NA       NA         NA          NA       NA
## 4 "AM WEST" on time          694    4840        383         320     201
## 5 ""      delayed          117     415         65         129      61
```

— Part 3

(3) Perform analysis to compare the arrival delays for the two airlines.

Compared percentage (not just counts) of either delays or arrival rates for two airlines overall. [Either with a chart, or a table; both would be exemplary]. Include text summarizes findings from this comparison. (+10)

Compared percentage (not just counts) of either delays or arrival rates for two airlines across five cities. [Either with a chart, or a table; both would be exemplary]. Include text summarizes findings from this comparison. (+10)

Describe discrepancy between comparing two airlines' flight performances city-by-city and overall. (+5)

Explain discrepancy between comparing two airlines' flight performances city-by-city and overall. (+5)

```
# The following values were calculated in Chunk 3

# AL_ontime
# AMW_ontime
# AL_delay
# AMW_delay
# AL_total
# AMW_total

data5 <- data3 %>%
  pivot_wider(
    names_from = Status,
    values_from = Arrivals
  )

data6 <- data5 %>%
  select(Airline, Cities)

data5 <- data5 %>%
  mutate(
    Total = (delayed + `on time`)
  )

data6 <- data5 %>%
  mutate(
    On_Time_Percentage = percent(`on time` / (delayed + `on time`)),
    Delayed_Percentage = percent(delayed / (delayed + `on time`))
  ) %>%
  select(Airline, Cities, On_Time_Percentage, Delayed_Percentage)
```

```
kable(data5, caption = "Delayed & On Time Arrivals for Alaska & AM West Airlines", align = "c")
```

Table 1: Delayed & On Time Arrivals for Alaska & AM West Airlines

Airline	Cities	on time	delayed	Total
ALASKA	Los Angeles	497	62	559
ALASKA	Phoenix	221	12	233
ALASKA	San Diego	212	20	232
ALASKA	San Francisco	503	102	605
ALASKA	Seattle	1841	305	2146
AM WEST	Los Angeles	694	117	811
AM WEST	Phoenix	4840	415	5255
AM WEST	San Diego	383	65	448
AM WEST	San Francisco	320	129	449
AM WEST	Seattle	201	61	262

```
kable(data6, caption = "Delayed & On Time Arrival Percentages for Alaska & AM West Airlines", align = "c")
```

Table 2: Delayed & On Time Arrival Percentages for Alaska & AM West Airlines

Airline	Cities	On Time Percentage	Delayed Percentage
ALASKA	Los Angeles	88.909%	11.091%
ALASKA	Phoenix	94.850%	5.150%
ALASKA	San Diego	91.379%	8.621%
ALASKA	San Francisco	83.140%	16.860%
ALASKA	Seattle	85.788%	14.212%
AM WEST	Los Angeles	85.573%	14.427%
AM WEST	Phoenix	92.103%	7.897%
AM WEST	San Diego	85.491%	14.509%
AM WEST	San Francisco	71.269%	28.731%
AM WEST	Seattle	76.718%	23.282%

—-Summary & Conclusion —-

AM West and Alaska Airlines both serve the Western United States, but from the arrival flight data it appears that their primary focus was on different cities. The majority of Alaska's flights were to Seattle (2146 arrivals) whereas AM West had the majority of arrivals in Phoenix (5255 arrivals).

Alaska Airlines saw an average delay percentage of 13% in flight arrivals whereas AM WEST saw a delay of 11% across all flights on the same five routes.

Alaska Airlines saw 501 delayed flights and 3274 on-time flights for a grand total of 3775 flights. AM West saw 787 delayed flights and 6438 on-time flights for a grand total of 7225 flights.

AM West was the only airline out of the two that had a delay percentage of total flights greater than 20% (in San Francisco and Seattle). However, these accounted for a smaller proportion of the overall number of flights so overall their performance was still better than Alaska airlines. Alaska, meanwhile, saw a 14.2%

delay on arrivals to Seattle but because Seattle accounted for over half of their arrivals, it reduced their overall performance and caused their overall on time percentage to be lower than AM West.

Both airlines saw their best performance in Phoenix, with a 5.1% delay percentage on arrivals for Alaska airlines and a 7.9% delay percentage for AM West.

The `pivot_wider()` and `pivot_longer()` functions were helpful in organizing the data for analysis. The exercise clarified the importance of structuring the dataframe in a way that makes analysis efficient by permitting operations across vectors.

—- Part 4

- (4) Your code should be in an R Markdown file, posted to rpubs.com, and should include narrative descriptions of your data cleanup work, analysis, and conclusions. Please include in your homework submission: The URL to the .Rmd file in your GitHub repository. and The URL for your rpubs.com web page.